Oral Cancer Detection Using AI/ML and Deep Learning

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Abstract—This paper describes the software design and implementation of a non-diagnostic prototype for binary oral image classification. Its PyTorch MergedNet model combines pooled visual features from VGG-16, ResNet-50, EfficientNet-B0, and MobileNetV2 into a 5,120-dimensional representation. The application includes a React frontend and a FastAPI backend; its inference code also contains test-time augmentation and Monte Carlo dropout uncertainty logic, with differences between the PyTorch and ONNX execution paths. The inspected repository contains 940 JPEG images arranged in cancer and non-cancer classes. A hash audit found 21 groups of exact duplicates encompassing 46 individual files (25 redundant copies), including 11 groups (25 files, 14 redundant copies) that cross data partitions. Patient identifiers are unavailable, and the dataset's original provenance and usage terms have not been verified. Although a model checkpoint is present, reproducible training records and empirical evaluation on the dataset were not found. Previously reported benchmark figures, including 99.92% accuracy, were generated synthetically and are not model test results. Accordingly, this paper makes no classification-performance or clinical-effectiveness claims. No evidence of regulatory clearance was found in the inspected repository, regulatory status has not been assessed, and the prototype is not clinically validated and is not intended for clinical use. It documents the prototype's implementation and its data, evaluation, and validation limitations.

Keywords—oral image classification, deep learning, feature fusion, uncertainty estimation, decision-support prototype.

I. INTRODUCTION

Oral cancer represents a notable global public health challenge. According to GLOBOCAN 2020 statistics, cancers of the lip and oral cavity accounted for an estimated 377,713 new cases and 177,757 deaths worldwide in 2020 [1]. Frontline screening strategies evaluated in a Cochrane systematic review center on conventional visual and tactile oral mucosal examinations; while evidence from a randomized trial suggested a reduction in oral cancer mortality in targeted high-risk individuals (tobacco or alcohol users), the review concluded that evidence to support screening programmes across the general population remains insufficient [2].

To explore software tools that could support frontline image review, automated photographic analysis using computer vision and deep learning has become an active area of medical-image analysis research [3], [4]. In this paper, the term "decision-support prototype" is defined specifically as terminology used in this paper to designate an exploratory software application intended to assist users in organizing, viewing, and reviewing photographic images, rather than as a universal clinical definition; the prototype is fundamentally not an autonomous diagnostic medical device. The prototype documented in this paper is strictly non-diagnostic: repository specifications and system requirements explicitly state that the software is not a substitute for biopsy, histopathology, or formal clinical evaluation by qualified healthcare specialists. No evidence of medical device clearance was found in the inspected repository, regulatory status has not been assessed, and the prototype is not clinically validated and is not intended for clinical use.

A central engineering design choice in visual image classification involves selecting an appropriate convolutional neural network (CNN) architecture. Individual convolutional backbones possess different architectural properties, receptive fields, and feature-extraction behaviors. For example, deep architectures with homogenous stacks of 3x3 convolutions (such as VGG-16) extract deep spatial feature hierarchies through increasing channel capacity across sequential pooling stages [5], residual networks (such as ResNet-50) incorporate residual connections that support the optimization and training of deeper network representations [6], and compact architectures (such as MobileNetV2 and EfficientNet-B0) optimize computational efficiency using depthwise separable convolutions with inverted residuals [7] and compound scaling across depth, width, and resolution [8]. Combining representations extracted across multiple diverse backbones—a technique referred to as feature fusion—aims to synthesize multi-scale representations into a unified feature vector prior to final classification.

In addition to classification, clinical imaging systems require mechanisms to communicate predictive confidence. Deterministic neural networks tend to output uncalibrated softmax scores that cannot reliably distinguish between certain and uncertain classifications. To address this challenge, Bayesian approximations such as Monte Carlo Dropout (MCD) [9] and data-perturbation methods such as Test-Time Augmentation (TTA) [10] have been proposed to quantify epistemic (model) and aleatoric (data) uncertainty, respectively.

However, applying deep learning to photographic medical datasets presents severe methodological and data-integrity challenges:
1) Image Redundancy and Partition Contamination: Photographic datasets often contain duplicate or near-identical images. If duplicate images appear across training and test splits, evaluation metrics measure memorization rather than true out-of-sample generalization [11], [12].
2) Input Quality Checking: The implementation includes a heuristic image-quality and blur-detection check; however, its effect on model behavior has not been evaluated.
3) Transparent Academic Reporting: Scientific rigor requires clear differentiation between verified software implementations, empirical performance measurements on a properly separated test set, and synthetic or heuristic software routines. In this repository, the current test partition is affected by exact duplicates and cannot be treated as an independent test set.

A. Objectives of This Paper

This paper presents an evidence-based system description of the oral image classification prototype based on a factual audit of the current repository. The specific objectives are: 1) To detail the system architecture of MergedNet, a multi-backbone feature fusion model implemented in PyTorch fusing feature vectors from VGG-16 [5], ResNet-50 [6], EfficientNet-B0 [8], and MobileNetV2 [7]. 2) To document the web application stack linking a React frontend client to an asynchronous FastAPI backend service. 3) To describe the inference-time uncertainty estimation routines present in code (Test-Time Augmentation and Monte Carlo Dropout) and detail the operational differences between the PyTorch and ONNX execution paths. 4) To disclose repository data limitations, including unverified dataset provenance, the presence of 21 duplicate image groups (with 11 groups crossing split boundaries), the absence of patient-level identifiers, missing reproducible training scripts, and the presence of synthetic benchmark generation scripts. 5) To articulate a conditional research roadmap outlining the technical and methodological prerequisites required before valid empirical performance claims can be evaluated.

II. RELATED WORK

Developing decision-support prototypes for oral mucosal image analysis draws upon research across computer vision, machine learning, and medical informatics. This section summarizes related literature concepts across four focused domains.

A. Prior Deep Learning Studies in Oral Image Classification

Convolutional neural networks have been investigated across several dedicated photographic oral lesion datasets. In an exploratory study, Welikala et al. [3] evaluated deep learning architectures, including ResNet-50, on their own curated dataset of intraoral photographs to classify lesions into suspicious and non-suspicious categories. In a separate institutional investigation, Warin et al. [4] trained and evaluated deep CNN architectures on their own dataset of 980 photographic images (comprising 365 OSCC, 315 OPMD, and 300 non-pathological images), reporting classification metrics specific to their dataset and experimental setup. While these studies demonstrate the algorithmic feasibility of training CNNs on specific image collections, they represent evaluations conducted strictly on their own distinct cohorts, and findings from prior research cohorts cannot be generalized to the unverified repository dataset or prototype evaluated in this project.

B. Feature Fusion Architecture

In this project, feature fusion refers specifically to the intermediate layer concatenation implemented in the MergedNet model. Global average pooled feature vectors extracted simultaneously from four distinct convolutional backbones—VGG-16 [5], ResNet-50 [6], EfficientNet-B0 [8], and MobileNetV2 [7]—are concatenated into a unified 5,120-dimensional representation that is fed directly into a common fully connected classification head.

C. Uncertainty Estimation in Deep Learning Systems

Standard deterministic deep neural networks output point estimates that do not represent prediction uncertainty [9]. In computational uncertainty literature, predictive uncertainty is broadly distinguished into two forms [9], [10]:

• Epistemic (Model) Uncertainty: Represents uncertainty in the network parameters arising from limited training data, which can theoretically be reduced with additional training samples. Gal and Ghahramani [9] formulated Monte Carlo Dropout (MCD) as a practical variational Bayesian approximation of deep Gaussian processes. By keeping dropout active at test time across repeated stochastic forward passes, the variance across the resulting predictive distribution provides a tractable estimate of epistemic model uncertainty [9].

• Aleatoric (Data) Uncertainty: Represents input-dependent uncertainty inherent to the observation process (such as image noise, optical artifacts, or blur), which cannot be eliminated simply by collecting more training data. Ayhan and Berens [10] investigated Test-Time Augmentation (TTA) in medical imaging, demonstrating that evaluating prediction dispersion across label-preserving geometric and optical perturbations can provide an empirical measure of data-dependent (aleatoric) uncertainty.

Importantly, while the prototype studied in this paper incorporates code for both Monte Carlo dropout and test-time augmentation in its inference pipeline, TTA is implemented in the codebase as an unweighted probability-averaging heuristic alongside an ad-hoc variance check. Neither method has been empirically calibrated or clinically validated as an uncertainty estimator on the repository's dataset.

D. Dataset Integrity, Partition Leakage, and Methodological Rigor

Systematic reviews and methodological surveys of machine learning literature have documented widespread vulnerabilities that undermine reported findings in published studies [11], [12]:

• Data Contamination via Duplicate Images: The repository audit identified exact duplicates across partitions. Exact duplicates could contaminate evaluation if the supplied training split was used to train the checkpoint [11], [12]. However, because the training history is missing, it cannot be confirmed whether the model actually saw those test images during training. In a literature survey across 17 quantitative fields, Kapoor and Narayanan [12] introduced an eight-part taxonomy of data leakage types, demonstrating that data leakage frequently produces overoptimistic performance conclusions.

• Absence of Patient-Level Partitioning: If multiple photographs collected from the same subject are divided randomly across training and testing sets, models can memorize patient-specific confounding features (such as surrounding dentition, skin pigmentation, or camera characteristics) rather than pathological indicators [11]. In their review of machine learning pitfalls in COVID-19 imaging, Roberts et al. [11] identified the failure to isolate patient records into disjoint splits as a primary source of bias and inflated reported accuracy.

• Missing Dataset Documentation in the Repository: In this repository, documentation regarding data provenance, original clinical sources, ethical approvals, and licensing terms is absent. In their study of COVID-19 imaging pitfalls, Roberts et al. [11] highlighted that missing source documentation and unclear dataset lineage impede scientific validation.

III. SYSTEM AND METHODS

This section documents the verified software architecture, repository dataset status, feature fusion model structure, and

uncertainty estimation mechanisms implemented in the prototype.

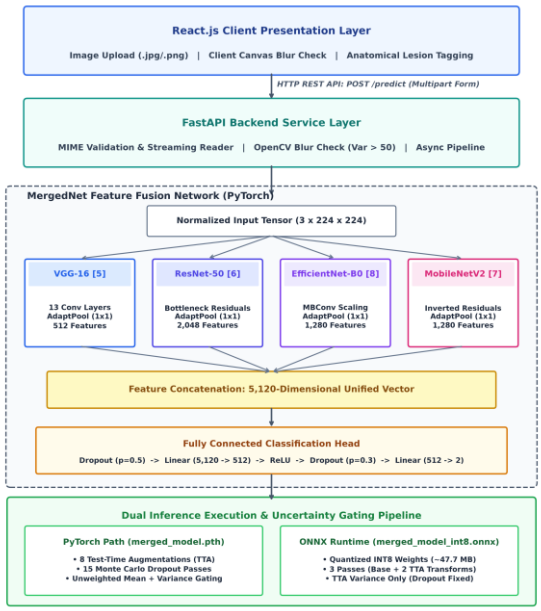


Fig. 1. End-to-end software architecture of the oral image classification prototype, illustrating the React frontend, FastAPI backend, MergedNet four-backbone feature fusion network (5,120-d concatenated representation), and inference-time uncertainty quantification pipeline.

A. System Architecture Overview

The application uses a decoupled client-server structure:
• Frontend Presentation Layer: Built with React.js, providing an interface for uploading oral images, executing client-side image blur checks, selecting anatomical locations, and viewing model output flags.
• Backend Service Layer: Implemented in Python using the FastAPI framework, handling REST API requests, file ingestion boundaries, image preprocessing, and inference orchestration.
• Inference Execution Engine: Executes model forward passes via PyTorch, with an alternative execution path using ONNX Runtime for CPU environments.

B. Dataset Characteristics and Inspection Findings

Inspection of the repository filesystem confirms an image collection arranged into folder partitions. Table I lists the verified image counts.

TABLE I. REPOSITORY IMAGE COUNTS BY PARTITION

Partition	Cancer Class	Non-Cancer Class	Total
Train	341	314	655
Validation	75	68	143
Test	74	68	142
Total	490	450	940

Audit findings regarding data integrity are summarized as follows:
1) Image Redundancy Audit: A SHA-256 cryptographic hash audit over all 940 images revealed 21 duplicate image groups encompassing 46 individual image files (representing 25 redundant copies across 21 unique images). Of these, 11 groups (encompassing 25 files, representing 14 redundant copies) crossed partition boundaries (e.g., identical images located simultaneously in train and test partitions). Consequently, the test partition cannot be considered independent from the training partition in its current state.
2) Patient-Level Identity: Filenames follow arbitrary numerical naming patterns (e.g., 0.jpg, 1.jpg). No patient identifiers, demographics, anatomical site metadata, or clinical labels are present in the repository. As a result, it is impossible to verify whether multiple photographs originated from the same subject.
3) Dataset Provenance and Licensing: Repository notes describe the image source as a Kaggle dataset; however, original source URLs, curation protocols, copyright licensing, and institutional ethical review records are absent from the codebase.

C. MergedNet Model Architecture

The core visual classification model, MergedNet, is defined in backend/models/merged_model.py. The architecture extracts features simultaneously across four parallel CNN backbones:
1) VGG-16 Backbone: Implements a 16-layer network [5] (13 convolutional layers arranged in five blocks with max pooling). The convolutional feature maps are pooled into a 512-dimensional vector.
2) ResNet-50 Backbone: Utilizes a 50-layer residual network [6] organized into four bottleneck stages. Global average pooling produces a 2,048-dimensional feature vector.
3) EfficientNet-B0 Backbone: Incorporates mobile inverted bottleneck (MBConv) stages with squeeze-and-excitation optimization [8], yielding a 1,280-dimensional feature vector.
4) MobileNetV2 Backbone: Employs the inverted residual convolutional structure [7] (mob.features), pooled into a 1,280-dimensional vector.
The pooled feature vectors are concatenated along the channel axis to form a composite 5,120-dimensional feature representation: 512 (VGG-16) + 2,048 (ResNet-50) + 1,280 (EfficientNet-B0) + 1,280 (MobileNetV2) = 5,120. This concatenated representation passes through a fully connected classification head consisting of:
• Dropout (p = 0.5) for regularization.
• Fully connected linear layer reducing dimension from 5,120 to 512.
• ReLU non-linear activation.
• Second Dropout layer (p = 0.3).
• Final linear output layer mapping 512 units to 2 class logits (Cancer vs. Non-Cancer).

D. Inference-Time Uncertainty Routines and Execution Differences

The application implements two runtime inference pathways:
1) PyTorch Pipeline: Executes model inference via PyTorch with uncertainty quantification logic:
• Test-Time Augmentation (TTA): Generates 8 augmented variants of the input image (rotations, flips, zoom, and brightness adjustments). Predictions across the 8 passes are averaged unweighted [10].
• Monte Carlo Dropout (MCD): Applies 15 stochastic forward passes with active dropout layers. The variance across passes is computed to measure predictive dispersion [9].
• Flagging Heuristic: If the maximum class probability variance across passes exceeds an empirical threshold (variance > 0.015), the system flags the prediction as 'uncertain'.
2) ONNX Runtime Pipeline: An alternative inference path

loads a quantized model (merged_model_int8.onnx) for CPU execution. In this path, Monte Carlo dropout cannot be activated because quantized ONNX graphs fix dropout behavior during graph compilation; consequently, the ONNX path computes prediction variance across only 3 passes (the base image plus two affine transforms), producing operational uncertainty differences between runtime environments.

IV. IMPLEMENTATION

The software application is built using a modern decoupled web architecture designed to facilitate oral image acquisition, processing, and review.

A. Backend Architecture

The backend is developed with Python 3.10 and FastAPI:
• Asynchronous Execution: Inference tasks run within an asynchronous worker pool (anyio.to_thread.run_sync) to prevent blocking the HTTP event loop during CPU/GPU tensor computation.
• File Ingestion Guardrails: The /predict endpoint enforces strict MIME-type validation, magic byte header verification, and streaming chunk size limits (max 10 MB) to prevent denial-of-service vulnerabilities.
• Server-Side Sharpness Filter: Evaluates image sharpness using the OpenCV Laplacian variance method (threshold = 50.0). If an image fails the sharpness check, the endpoint returns a blur advisory flag alongside prediction outputs.

B. Frontend Architecture

The frontend is implemented as a React.js single-page application:
• Client-Side Blur Screening: A client-side JavaScript utility calculates image gradient variances on an HTML5 canvas to provide a preliminary blur warning before network transmission.
• Anatomical Tagging: An interactive SVG-based oral cavity map allows users to designate the anatomical region of the photographed lesion (e.g., buccal mucosa, lateral tongue, gingiva).
• Status Flag Presentation: The user interface displays categorical output badges, blur advisories, and uncertainty flags.

C. Differentiation of Machine Learning vs. Heuristic Components

To ensure accurate technical reporting, Table II categorizes all analytical modules present in the repository, differentiating neural network components from rule-based and heuristic algorithms.

TABLE II. CATEGORIZATION OF REPOSITORY SOFTWARE COMPONENTS

System Component (Code Location)	Implemented Mechanism	System Category
Oral Image Classification (ml_engine.py)	PyTorch MergedNet: 4-backbone feature fusion [5]–[8]	Neural Network Checkpoint
Image Sharpness Screening (image_quality.py)	OpenCV Laplacian variance check (Threshold = 50.0)	Deterministic Heuristic
Specular Glare Suppression (optical_normalization.py)	Luminance threshold mask + OpenCV inpainting	Classical Computer Vision
Color Normalization (optical_normalization.py)	Reinhard statistical mean/std transfer in LAB color space	Statistical Color Transfer
Boundary Segmentation (vision_pipeline.py)	Erythema index (R-G)/(R+G) + brightness thresholding	Handcrafted Heuristic
Epidemiological Risk Scoring (clinical_risk.py)	Linear sum: Age (+0.3), Tobacco (+0.35), Alcohol (+0.2)	Rule-Based Formula
Four-Class Ordinal Triage (multimodal_fusion.py)	Probability modification with bonus weights and T = 2.5	Rule-Based Formula
AJCC Clinical Staging (clinical_staging.py)	Decision tree mapping lesion diameter and site to stage	Rule-Based Logic
Grad-CAM++ Explainability (xai_cam.py)	Generalized gradient activation maps on ResNet-50 [13]	Post-Hoc Feature Map

Note on Latent Cross-Attention Code: The file app/services/multimodal_fusion.py contains a PyTorch class definition named MultimodalCrossAttentionFusion. However, live code inspection confirms that this neural network module is not invoked during /predict request handling; instead, the live endpoint executes the rule-based function evaluate_multimodal_ordinal_triage.

D. Model Checkpoint Status and Missing Training History

The repository includes two compiled model files: merged_model.pth (PyTorch weight dictionary, ~94.7 MB) and merged_model_int8.onnx (Quantized ONNX model, ~47.7 MB). While these files load into the MergedNet model definition, no training scripts, optimization configurations, or training logs are present in the repository. Essential training parameters—such as the optimizer (e.g., AdamW, SGD), learning rate, learning rate schedule, loss function formulation, batch size, epoch count, and data augmentation policies used during training—remain undocumented. Furthermore, training and validation loss curves are absent from the codebase.

V. EVALUATION STATUS AND LIMITATIONS

Transparent reporting requires full disclosure of how performance figures were generated and what evaluation data currently exists.

A. Disclosure of Synthetic Benchmark Generation Scripts

Internal project documents and prior drafts cited high classification figures, including an overall accuracy of 99.92%, alongside comparative baseline tables and ablation studies.

Code inspection of the benchmark script (backend/scripts/ieee_benchmark_evaluator.py) reveals that these reported figures are synthetic artifacts generated by random numerical simulation routines. Specifically:

- The script implements a simulation function (run_comparative_benchmark) that generates synthetic sample sizes (N = 1,200) and draws prediction probabilities from parameterized statistical distributions (such as normal and beta random distributions).
- Performance metrics for baseline architectures (standalone ResNet-50, VGG-16, EfficientNet-B0, MobileNetV2) and ablation variants were generated by computing summary statistics over these synthetic random values.
- The script automatically exports these values into JSON records (ieee_benchmark_results.json) and formatted LaTeX tables (ieee_tables.tex).

These synthetic outputs represent an illustrative simulation script written during software development; they do not represent empirical evaluations of the model on real images. These figures are strictly excluded from this paper.

B. Absence of Reproducible Empirical Model Evaluation

Inspection of the repository found no reproducible, verifiable empirical evaluation of the saved model weights (merged_model.pth) across the repository's image collection. Consequently:

- Real test-set performance metrics—including Classification Accuracy, Sensitivity, Specificity, Positive Predictive Value, Negative Predictive Value, F1-Score, and AUROC—are currently unavailable in existing project records.
- No claims of high classification accuracy, superior performance over baseline models, or diagnostic capability are made in this paper.

C. Data Integrity and Partition Leakage Limitations

The dataset presents several data integrity issues that preclude valid scientific conclusions:

1) Duplicate Images Across Partitions: The hash audit identified 21 exact duplicate image groups in total (46 individual files, 25 redundant copies), with 11 groups (25 files, 14 redundant copies) crossing partition boundaries (appearing in multiple split folders, including train and test). If the checkpoint was trained on the supplied training partition, evaluating it on this test partition could measure memorization rather than generalization [11], [12]; however, because training records are absent, actual sample exposure during training remains unverified.

2) Lack of Patient-Level Separation: In the absence of patient identifiers, it cannot be verified whether multiple photographs from the same individual exist across partitions. In biomedical machine learning literature, patient overlap between training and test sets has been documented as a frequent cause of performance overestimation [11], [12].

3) Small Sample Size: A total cohort of 940 images (490 cancer, 450 non-cancer) is modest for training a model architecture with 5,120 fused latent dimensions, making overfitting a possible concern that has not been evaluated.

D. Ethical, Regulatory, and Clinical Boundaries

1) No Ethics Documentation Found: No documentation regarding Institutional Review Board (IRB) ethics approval, human subject protocols, or patient consent forms was found in the repository.

2) Absence of Clinical Validation: The prototype has not undergone prospective clinical trials, blinded reader studies with qualified clinicians, or validation on external cohorts.

3) Regulatory Status and Clinical Intent: No evidence of medical device clearance or regulatory authorization (such as FDA, CE-IVD, or CDSCO approval) was found in the inspected repository, and regulatory status has not been assessed. The prototype is not clinically validated and is not intended for clinical use.

VI. DISCUSSION

A. Technical and Engineering Scope

The primary contribution of this work is the software engineering design and architectural implementation of a multi-backbone decision-support prototype. From a systems perspective:

- Multi-Backbone Architecture: MergedNet demonstrates an operational implementation for extracting and concatenating multi-scale visual features from four distinct CNN backbones [5]-[8] into a single classification head.
- Uncertainty Quantification in Code: The inclusion of Test-Time Augmentation [10] and Monte Carlo Dropout [9] illustrates how uncertainty estimation logic can be embedded into decision-support inference endpoints.
- Decoupled Web Architecture: The separation of a React client from an asynchronous, memory-bounded FastAPI backend demonstrates a practical architecture for serving deep learning models over HTTP.

B. Methodological Lessons

The audit of this undergraduate capstone project provides valuable methodological lessons for applied artificial intelligence research:

1) The Necessity of Rigorous Data Auditing: Performing cryptographic hash audits and verifying patient-level splits are essential first steps before conducting model training or evaluation [11], [12]. Data contamination across partitions invalidates reported test results.

2) Transparent Documentation: Distinguishing between mock simulation scripts (such as the synthetic benchmark evaluator) and genuine empirical evaluation on real data is vital for academic integrity.

3) Clear Component Demarcation: Auxiliary clinical tools—such as risk questionnaires and staging calculators—should be clearly documented as rule-based heuristics rather than conflated with trained machine learning models.

C. Conditional Remediation and Research Roadmap

Before valid empirical performance claims or further research exploration can be considered, the following conditional steps must be undertaken:

- Phase 1: Dataset Provenance and Permissions (Conditional): Trace the original source of the Kaggle dataset and verify whether ethical approvals, patient consent, and research-use permissions permit academic publication.
- Phase 2: Deduplication and Patient-Level Splitting (Conditional): Remove exact SHA-256 duplicate image pairs. If patient-level metadata is obtainable, isolate subject records into mutually disjoint partitions; otherwise, document the lack

of patient-level separation as a primary limitation.
• Phase 3: Reproducible Training Pipeline: Implement a version-controlled PyTorch training script (train.py) with explicit loss formulation, optimizer, learning rate schedule, and logged training/validation convergence curves.
• Phase 4: Empirical Evaluation and Ablation Analysis: Retrain and evaluate the architecture on an independent test partition, reporting empirical Accuracy, Sensitivity, Specificity, F1, and AUROC, with ablation studies comparing MergedNet to single backbones.
• Phase 5: Refinement of Auxiliary Modules (Conditional): Explore replacing heuristic color segmentation with validated deep learning models and re-evaluating risk scoring formulas against clinical evidence.
• Phase 6: Clinical Collaboration (Conditional): Following formal ethical clearance and completion of technical milestones, partner with clinical specialists to formulate prospective validation protocols.

VII. CONCLUSION

This paper has documented the software design and implementation of a non-diagnostic prototype for binary oral image classification. The engineering framework implements MergedNet, a multi-backbone convolutional neural network combining pooled visual features from VGG-16 [5], ResNet-50 [6], EfficientNet-B0 [8], and MobileNetV2 [7] into a 5,120-dimensional representation, served through a FastAPI backend and a React frontend with inference-time uncertainty logic [9], [10].

Rather than asserting unverified performance figures, this paper provides an honest, evidence-based account of the prototype's current implementation and limitations:
• The prototype is not clinically validated, is not intended for clinical use, and no evidence of regulatory clearance was found in the inspected repository (regulatory status has not been assessed).
• The dataset comprises 940 JPEG images (490 cancer, 450 non-cancer) with unverified provenance and no patient-level identifiers.
• A hash audit identified 21 duplicate image groups encompassing 46 files (11 groups encompassing 25 files crossing partition boundaries), precluding the current test partition from serving as an independent test set.
• Training scripts and logs are absent, and previously reported benchmark metrics (e.g., 99.92% accuracy) were generated synthetically rather than via empirical evaluation.
• Genuine empirical evaluation results on the dataset are unavailable.

By prioritizing scientific integrity, data transparency, and objective engineering documentation over premature performance assertions, this paper provides a transparent baseline for future academic work.

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